

— SNAPDRAGON AI LAB BUILD & PRESENT CHALLENGE 2026

Sahaay

सहाय · assistance

Hands-free and eyes-free control of a
Windows PC, **running entirely on the
Snapdragon Hexagon NPU.**

Head and voice replace the mouse and keyboard; a conversational narrator
replaces the screen. Six AI models, no cloud, no data leaves the laptop.

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BUILT FOR SNAPDRAGON-POWERED HP PCS

Two groups of people who cannot use a laptop the way it ships.

MOTOR IMPAIRMENT

Cannot use a mouse or a keyboard

Cerebral palsy, spinal injury, ALS, severe RSI, a broken arm. **The screen works; the hands do not.**

Today: every assignment, every portal, every chat needs someone else's hands, or a ₹1.5 lakh eye-gaze system.



LOW VISION AND BLINDNESS

Cannot see the screen

Diabetic retinopathy, glaucoma, congenital blindness. **A screen reader lists forty buttons; it cannot describe a photo, a chart, a scanned form, or the room.**

Today: anything that is an image, a canvas or the physical world needs someone else's eyes.



2.68

crore

Indians living with a disability
Census of India 2011

1 in 5

of them with a movement disability
Census of India 2011

1 in 5

of them with a visual disability
Census of India 2011

Assistive technology today is expensive, depends on the cloud, and stops at the basics.

COST

₹1.5 lakh+

for an eye-gaze system

Plus a licence, a dongle, and often a subscription. Out of reach for a student on a ₹60,000 laptop budget, and rarely covered by insurance in India.

Tobii Dynavox, PCEye and comparable Indian resellers, 2026 list prices

PRIVACY

24 h/day

of face video and screen content

A cloud assistant has to stream the camera and the screen to a server. Face data is **sensitive biometric data under India's DPDP Act 2023**; the screen shows bank statements and medical records.

Digital Personal Data Protection Act 2023, s.2(t) and s.9; typical cloud speech and vision APIs

COVERAGE

0

built-in tools that describe a photo or the room

Screen readers read controls, not images. Voice control types and clicks, but still needs a hand for everything else. **Nothing ties head, voice, screen and camera together.**

Windows Narrator and Voice Access feature scope, Microsoft docs, 2026

Run the entire assistive stack on the NPU that is already inside the laptop.

A Snapdragon X laptop ships with a 45 TOPS Hexagon NPU (80 on X2). It can run face tracking, speech recognition, a language model and a vision model at the same time, with camera frames, audio and screenshots never leaving RAM. HP sells that laptop to students as the OmniBook.

₹1.5 L+

eye-gaze hardware, licence, cloud account

₹0

Sahaay: a download for the Snapdragon laptop you already own



Two modes, one on-device brain: hands-free control and a screen narrator.

HANDS-FREE MODE

Head moves the cursor. Voice types and commands.

For motor impairment, injury, RSI

- **Head cursor:** nudge past a dead zone and the cursor moves; hold still and it stays. Calibrates itself in one second.
- **Clicks without hands:** short blink = left click, long blink = right click, dwell 0.9 s = click, mouth open = drag.
- **Dictation** into any application, English and Hindi.
- **25 instant commands** (click, scroll, press ctrl+s, next window, open Chrome) in under a millisecond.
- **Free-form commands** through the language model: "open my downloads and sort by date"
- **Physical switches:** any Arduino becomes a click and push-to-talk interface.

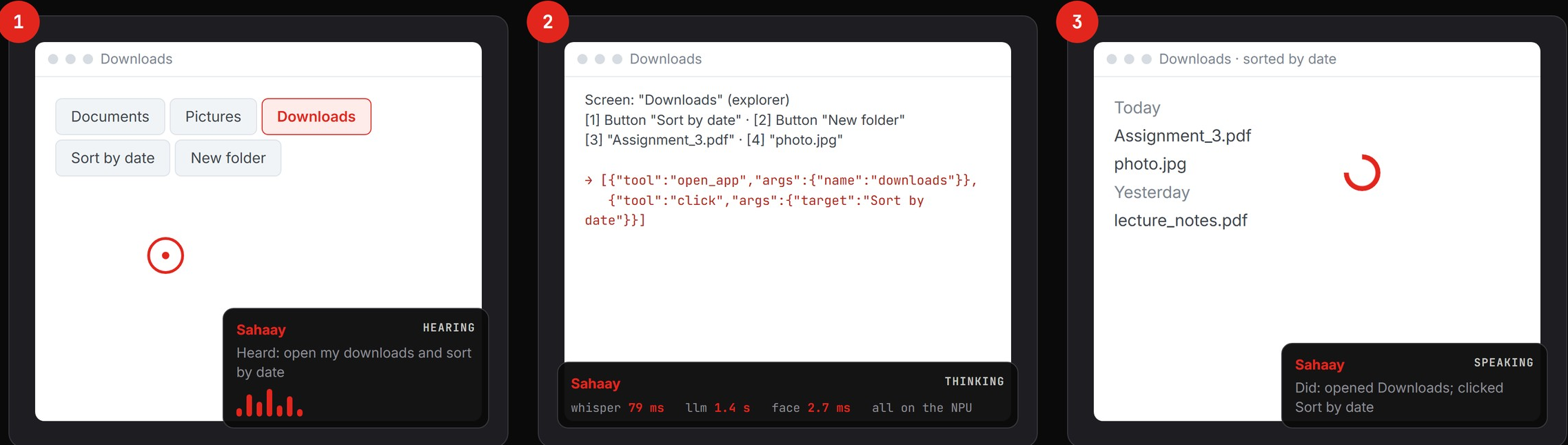
NARRATOR MODE

The screen and the camera, described in speech.

For low vision and blindness

- "what's on my screen" the UI Automation tree summarised in two sentences: which app, what content, what matters.
- "read this" the focused paragraph or field, read aloud.
- "describe the picture" a vision-language model explains photos, charts, canvases, scanned forms.
- "what do you see" the webcam view: people, objects, labels, signs.
- **Every action is confirmed by voice**, so the user always knows what just happened.
- **English and Hindi voices**, synthesised on the laptop.

How a spoken request becomes a validated action and a spoken confirmation.



THE USER SAYS

"Open my downloads and sort by date."

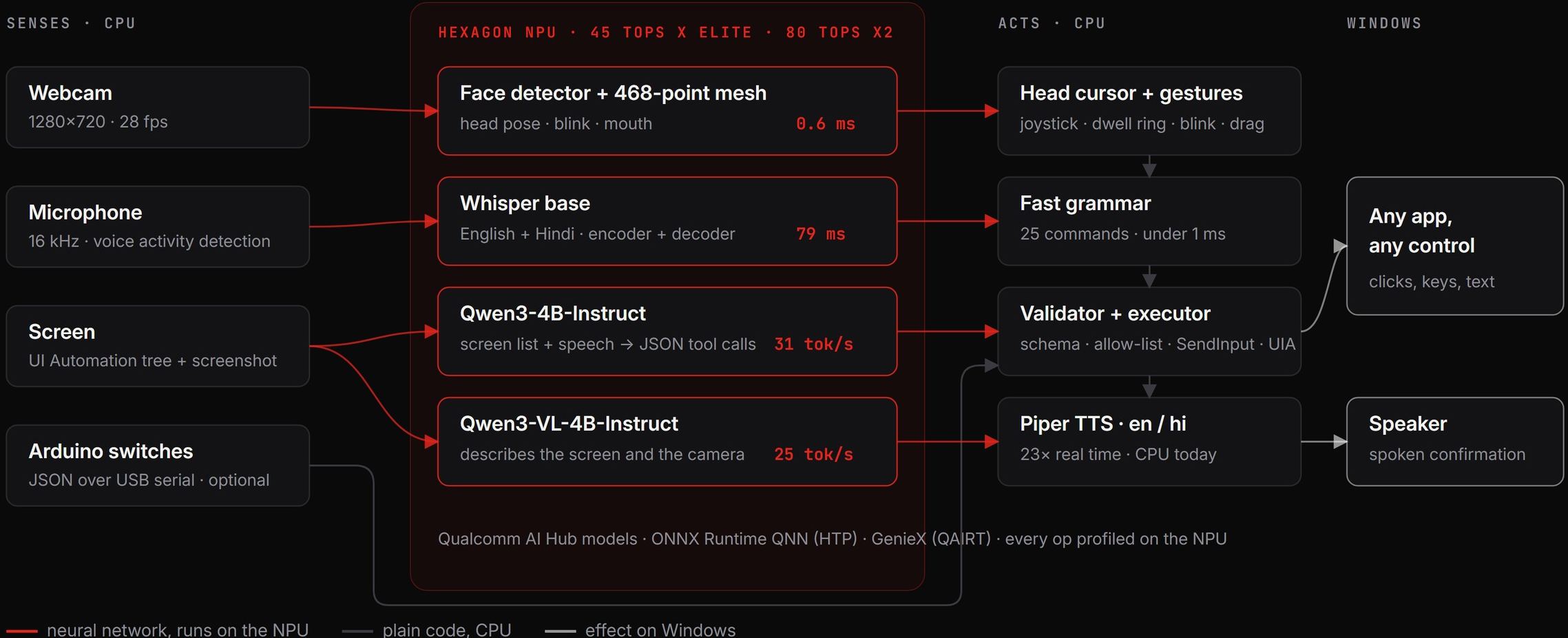
ON THE NPU

Whisper transcribes. UI Automation lists the controls. Qwen3-4B returns two tool calls; the validator checks both against the allow-list.

SAHAAY SAYS

"Downloads is open, sorted by date. Newest first is Assignment 3."

Six models on the Hexagon NPU. Capture, validation and input injection on the CPU. The GPU untouched.



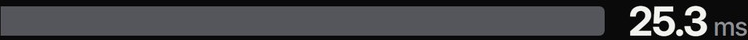
Every number below is measured, and the NPU execution provider is asserted before it is reported.

Latency per stage, CPU vs NPU

Median of 30 runs on the development laptop (Snapdragon X2 Elite). Same graph, same inputs.

Face pipeline

detect, crop, 468-pt mesh,
pose · live 720p frame



Face detector

BlazeFace 256×256



Face mesh

468 landmarks 192×192



Oryon CPU Hexagon NPU via QNN bars scaled per row

`tools/npu_check.py` asserts the active provider for every model. The usual `providers=[...]` call silently runs on the CPU with today's runtime; Sahaay selects the NPU device explicitly.

Speech and generative models, NPU only

Whisper ships as a precompiled QNN context; the language models run through GenieX on QAIRT.

79_{ms}

to transcribe a 5.6 s command with
Whisper base (21 ms encoder, 2.7 ms per
token)

NPU

31_{tok/s}

Qwen3-4B planning tool calls; 243 ms to
first token with the screen in context

NPU

25_{tok/s}

Qwen3-VL-4B describing the screen or
the webcam; 269 ms to first token

NPU

10_s

to load each generative model; face
tracking is up in 2 s, so the cursor works
almost immediately

NPU

QUALCOMM AI HUB, REFERENCE DEVICES, EVERY OP ON THE NPU

X2 Elite	face 0.4 ms · mesh 0.2 ms · Whisper enc 21.5 ms · dec 2.4 ms/token
X Elite (HP OmniBook)	face 0.7 ms · mesh 0.3 ms · Whisper enc 45.4 ms · dec 3.7 ms/token

One command installs it on any Snapdragon X or X2 PC. After that it never needs the internet.

```
# HP OmniBook, EliteBook, any Snapdragon X / X2 PC · no admin rights
powershell -ExecutionPolicy Bypass -File install.ps1
# Python if missing · venv · models for your chipset, once · espeak-ng · Start Menu shortcut
```

PRIVATE

No account, no cloud, no telemetry

Camera frames, audio and screenshots are processed in RAM and discarded. Zero network calls at runtime; verifiable by running with Wi-Fi off. A local log file for support, nothing else written.

FITS THE JUDGE'S LAPTOP

16 GB for hands-free mode, 32 GB for everything

Measured process growth: language model **5.4 GB**, vision model **6.1 GB**. On a 16 GB OmniBook 3 or 5 run with `--no-vlm` or the 2B vision model; on a 32 GB OmniBook X or Ultra load both. 45 TOPS is enough.

ACCESSIBLE

Usable without sight or hands

Global hotkeys, a tray icon, a heads-up panel showing what was heard and done with live per-model latency, spoken confirmation of every action, English and Hindi voices, and switch access through any Arduino, including the UNO Q from the AI Lab kit.

Where the cloud fails this user, and the trade-offs Sahaay accepts in return.

WHAT THE USER NEEDS	CLOUD ASSISTANT	SAHAAY ON THE HEXAGON NPU
Privacy of a face camera and the screen	Frames and screenshots leave the device continuously. Face data is sensitive biometric data under India's DPDP Act 2023 (s.2(t)); the screen carries bank and medical records	Processed in RAM, discarded. zero network calls after install; verifiable with Wi-Fi off
Cursor latency a person can tolerate	A network round trip before any inference. typically 50 to 150 ms to a cloud region from India, more on congested mobile links	2.7 ms per frame for the cursor; 79 ms for a spoken command. measured, benchmarks/results.md
Works in a hostel, a hospital ward, a train	Stops when the link drops. every command is a request; there is no offline mode	Fully offline. 6 GB of models cached once; nothing else fetched
Battery on a laptop that is the only interface	Video upload keeps the radio busy; inference still runs somewhere. continuous 720p upload is 1 to 3 Mbps over Wi-Fi or 4G	The CPU path would cost 25 ms per frame; the NPU does it in 2.7 ms and the CPU stays idle for the user's apps. measured, same frame
Cost to a student	Per-minute billing for speech, plus vision and language calls. Whisper API at \$0.006/min: three hours a day is about ₹2,700 a month, speech alone (₹84/\$)	Zero after the laptop. no subscription, no API keys

TRADE-OFFS WE ACCEPT **4-billion-parameter models** reason less well than frontier cloud models, so common commands bypass them through a fixed grammar and every model action is validated. **Each generative model takes 10 s to load** and 5 to 6 GB of RAM. **Whisper base mishears in noisy rooms**; fragments are dropped rather than acted on.

Seven problems between the models and a working product, and their fixes.

01 ONNX Runtime silently ran on the CPU while claiming the NPU

With the current plugin packaging, `providers=["QNNExecutionProvider"]` is accepted and ignored. **Fix:** register the QNN plugin, pick the NPU device through the device API, assert the active provider on every session, and ship `npu_check.py` so a reviewer can see it.

02 No OpenCV for Windows on ARM64

The camera and every image operation had to work without it. **Fix:** DirectShow through COM for capture (with `CoInitialize` on every worker thread), Pillow and numpy for the MediaPipe letterbox, affine crop and landmark mapping.

03 Whisper ships as a precompiled QNN context with no Python runner

Qualcomm's reference decode loop needs torch. **Fix:** re-implemented the encoder call and the 200-step KV-cache decoder loop in numpy: six layers of self-attention cache, causal mask and forced language prefix.

04 Piper's phonemiser is not built for ARM64

The wheel's espeak bridge fails to import. **Fix:** phonemise through the espeak-ng CLI and drive the VITS ONNX graph directly; English and Hindi voices at 23× real time.

05 Two GenieX models cannot initialise at the same time

Concurrent loads fail to create HTP contexts. **Fix:** a single loader thread brings up the language model, then the vision model, while the cursor and speech are already live.

06 A mis-heard fragment launched Windows Search

"Open it." from clipped audio became a search for "it". **Fix:** pronoun and length guards in the grammar, an allow-list of tools and elements, and no search fallback for short tokens.

07 QNN context caching and chipset generations

A cached compiled graph must be reused, not regenerated, and the development laptop's Hexagon v81 binaries do not run on the OmniBook's v73. **Fix:** reuse the cache when present, invalidate on mismatch, and ship separate model packs per chipset, both profiled on Qualcomm AI Hub.

What ships next, in order.

- **Gaze on the NPU.** The AI Hub eyegaze model for coarse screen-region selection alongside head pose, for people with very limited neck movement.
- **Piper on the NPU, Hindi and Hinglish commands.** The AI Hub Piper recipe, and a command grammar that understands "neechе scroll karo".
- **Element grounding from the vision model.** "Click the blue button" when an app has no accessibility tree: games, canvases, remote desktops.
- **Trials with an assistive-technology centre.** Calibration profiles for tremor and limited range.
- **MSIX package** for the Microsoft Store, and OmniBook presence-sensor auto-pause.

A ₹60,000 laptop that anyone can use, because the NPU is already inside it.

Everything in this deck is in the repository: code, benchmarks, installer, model packs for both chipsets, and the reviewer tools that prove the NPU is doing the work.

github.com/Vision-jarvis/sahaay

Ruhan Srivastava

IIT Kharagpur · built on a Snapdragon X2 Elite with Qualcomm AI Hub, ONNX Runtime QNN and GenieX

